

Rheological properties of SBS-modified asphalt containing aluminum hydroxide and organic montmorillonite

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ABSTRACT

The composite flame retardant composed of aluminum hydroxide (ATH) and organic montmorillonite (OMMT) is of potential application value in the tunnel asphalt pavement. To investigate the effect of ATH and OMMT on the rheological properties of SBS modified asphalt at high and low temperatures. In this study, the composite ATH/OMMT/SBS modified asphalt was subjected to multiple stress creep recovery (MSCR), linear amplitude sweep test (LAS), and Bending beam rheometer (BBR) experiments. Analysis of high and low temperature rheological properties and fatigue properties of composite ATH/OMMT/SBS modified asphalt under the conditions of single incorporation and composite modification. The findings of the tests suggest that ATH can significantly improve the fatigue life but has adverse impact on the creep recovery property of asphalt materials, OMMT has a significant improvement effect on the low-temperature performance, however, the detrimental effect of OMMT on fatigue performance is greater than the improvement of ATH. The research results provide a reference for the practical application of flame retardants.

1. Introduction

1.1. Review

Asphalt pavement is widely used in tunnel pavement laying due to its good road performance. However, the flammability of asphalt materials gives the tunnel asphalt pavement potential safety risks in the process of vehicle operation [1]. In detail, the combustion of asphalt materials in a semi-closed space of the tunnel type will release massive heat and toxic gases, threatening driving safety and hindering rescue work. To prevent the tunnel asphalt pavement from burning, previous studies usually focused on the structure and material [2,3] and reached the goal by installing the porous asphalt pavement with the function of emptying flammable liquids on the road surface or by developing the flame retardant asphalt. However, the former's flame retardant efficiency depends on the connected state of porous pavement holes that are prone to blockage [4]. At the same time, the adhesion problem is easily generated by the asphalt mixture in the humid tunnel environment [5,6]. These further limits the application of such a method. Moreover, the preparation of flame retardant asphalt by adding fire retardants to asphalt materials is now one of the most commonly used flame retardant strategies in the tunnel asphalt pavement.

The flame retardants commonly used include organic, inorganic and compound flame retardants. Among them, some organic flame

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retardant agents (such as halogen) are gradually replaced by inorganic flame retardants due to their toxicity [7]. Inorganic flame retardants represented by aluminum hydroxide (ATH) have the advantages of fire prevention, smoke suppression, money-saving and environmental protection [8]. As the research and development basis of many composite flame retardants, the research and application of ATH - based composite flame retardants has always been a hot spot in the research of flame retardant asphalt pavement. To investigate the flame retardant performance of the ATH-modified asphalt on a macro and micro level, Yang et al. [2] used limit oxygen index (LOI), thermogravimetry-Fourier transform infrared spectroscopy (TG-FTIR) and scanning electron mirror (SEM) test. They pointed out that ATH mainly generates a large amount of water vapor through endothermic degradation, diluting combustible or toxic gases, and generates carbon layer structure (Al_2O_3) that can block air and heat transmission on the hot surface of asphalt materials. However, in order to achieve good barrier effect, ATH is generally used with a high dosage (more than 20 %) [9], which adversely affects the mechanical properties of asphalt materials and further reduces the service life of asphalt pavement. At the same time, the performance of asphalt pavement is significantly affected by asphalt aging [10], and the rheological index is an important index to evaluate the aging resistance of asphalt binder. Short-term thermo-oxidative aging of asphalt mixtures occurs during production and paving, which is mainly due to oxidation and loss of volatile components at high temperatures [11]. The OMMT modified asphalt not only has the unique advantages of reducing flammability and improving thermal stability, but also can improve the high temperature performance of SBS modified asphalt [12]. Liu et al. [13] studied the rheological properties of modified asphalt with different OMMT contents through microscopic testing methods such as Fourier transform infrared spectroscopy (FTIR) testing and fluorescence microscopy (FM) testing. The results show that the addition of OMMT to the prepared warm mix asphalt can significantly optimize its fatigue performance, enhance the storage stability and UV aging resistance of the asphalt, and have little effect on its low temperature performance. L. Haurie [14] showed that although OMMT had some effect on slowing down the pyrolysis of polymers, the values of limiting oxygen index and other flame retardancy tests did not improve in the test. Therefore, the use of OMMT as a unique flame-retardant material is limited due to its low efficiency. However, due to the possible synergistic effect of OMMT with metal hydroxide flame retardants, it has been further studied. Zhu et al. [15] studied the flame retardant characteristics of ATH- organic montmorillonite (OMMT) modified asphalt through LOI, conical calorimetry and thermogravimetry test. They believed that the method of combining OMMT with ATH could not only reduce the ATH content, but also improve the flame retardant performance of asphalt materials. For the flame retardant mechanism of both, scholars [16] pointed out that the mesh structure formed by mixing OMMT improves the thermal stability of the asphalt matrix. Moreover, the existence of winding path effects [17] extends the volatile path of combustible/toxic gases. Under the dual function of volatile gas and water vapor, OMMT gradually migrates and enriches the thermal surface of asphalt materials, forming a dense carbon layer structure with Al_2O_3 to hinder the combustion reaction. Therefore, there is a research foundation for the flame-retardant effects and mechanisms of ATH and OMMT in asphalt materials. Asphalt materials prepared through synergistic modification with ATH and OMMT not only possess excellent flame-retardant and smoke-suppressing properties, but also exhibit significant advantages in terms of environmental protection, economic benefits, and more.

1.2. Scope and objectives

However, there is currently limited research on the flame retardant and smoke suppression performance of ATH/OMMT composite modified asphalt in tunnels, and there has been insufficient exploration into the rheological properties of ATH/OMMT composite

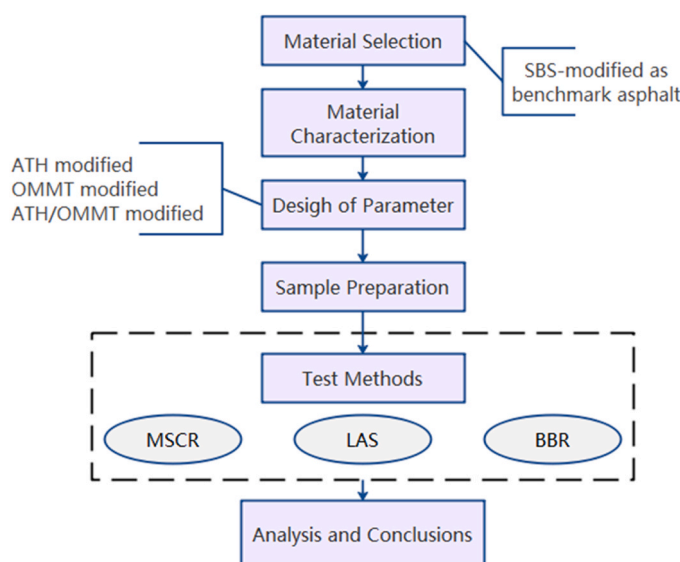


Fig. 1. Research methodology.

modified asphalt. This lack of research hinders the advancement of ATH/OMMT modified asphalt for application in tunnel asphalt pavement. Therefore, in this paper, ATH and OMMT are used as modifiers, and SBS-modified asphalt, which is commonly used in tunnel pavement, is used as the reference asphalt to prepare flame-retardant asphalt. The comparison was made with single-mixed, compound-modified and unmodified asphalt. By testing its high-temperature rheological properties (MSCR), medium-temperature fatigue properties (LAS) and low-temperature crack resistance (BBR), the influence of ATH/OMMT composite modified asphalt on pavement rheological properties was systematically analyzed. The overall research methodology of the current study is presented by a flow chart in Fig. 1. This study can provide reference for researching tunnel fire-retardant and smoke-suppressing materials with good mechanical properties.

2. Materials and methods

2.1. Materials

With the increasing vehicle load and traffic flow, ordinary asphalt pavement is difficult to meet the service requirements of roads in most parts of China [18]. Especially under the extreme climate environment, many diseases appear on ordinary asphalt pavement soon after it is opened to traffic. The styrene-butadiene-styrene (SBS) modified asphalt not only has good high temperature performance and low temperature crack resistance, but also can significantly improve the water stability of asphalt pavement. Due to these excellent performances, SBS modified asphalt plays a leading role in highway and tunnel asphalt pavement in China in recent years [17]. In this study, SBS modified asphalt was used as the control asphalt.

The selected SBS modified asphalt was produced by high speed-shear mixing of SK90# base asphalt and 5 wt% SBS modifier. The shear temperature was 180°C. The main technical indexes are shown in Table 1.

ATH was selected as asphalt modifier, in which ATH is white crystal powder, specific gravity is 2.46 g/cm³, and its main physical and chemical index is shown in Table 2.

The commonly used organic montmorillonite modifier (OMMT) is selected, in which the MMT content is greater than 95 %, and its specific gravity is 1.68 g/cm³. Fig. 2(a) and (b) are the material images of ATH and OMMT.

The chemical composition of OMMT was mentioned in our earlier study. Fig. 3(a) and (b) are the SEM images of ATH and OMMT, respectively. It can be seen that the particle size of ATH and OMMT are basically less than 10 µm, which belong to nano-scale materials. There is no obvious agglomeration or agglomeration between them, but the OMMT modifier has obvious lamellar structure. In the early stage, the penetration, softening point, ductility and limiting oxygen index (LOI) of the composite modified asphalt were tested and analyzed, and the preferred content of ATH was 10 %, and the content of OMMT modifier was selected as 1 % and 3 %.

2.2. Sample preparation

The production process of all modified asphalt was the same, and the specific operations followed the steps shown below:

- (1) The modifier was placed in the oven at 105°C for 4 h until moisture was completely evaporated; heating the SBS modified asphalt to the melting state (170–180°C), and poured 500 g of molten asphalt into the reaction tank;
- (2) The reaction tank with hot asphalt was placed in a constant temperature magnetic heating agitator to keep a constant temperature (180°C), adding the specified mass OMMT and ATH slowly, and stir manually.
- (3) Start the high-speed shearing machine shearing asphalt for 45 min at 5000 rpm speed, and then adjust it to the low speed (1000 rpm) to shear 10 min to remove the bubbles in the asphalt;
- (4) After the preparation of the modified asphalt was completed, take out the reaction tank, pour the prepared modified asphalt into a special aluminum box and seal it for storage, and test it after cooling.

2.3. Test methods

2.3.1. Multiple stress creep recovery (MSCR) test

SHRP in the United States proposed the use of "rut factor" index to characterize the high temperature performance of asphalt materials [19], but the correlation between rut factor and the actual rutting resistance performance of pavement is poor [20]. The main reason is that the loading mode of the index in the test process cannot effectively simulate the load-deformation response model of the real pavement system. Rutting factor is a material parameter within the range of linear viscoelasticity, reflecting the mechanical response of asphalt material in the non-damage state, and the deformation of asphalt material at high temperature should be a process of generating and accumulating permanent deformation in the damaged state [21].

Table 1

Technical indicators of the SBS modified asphalt binder.

Penetration (25°C, 100 g, 5 s)/0.1 mm	Ductility (5 cm/min, 5°C)/cm	Softening point /°C	RTFOT (163°C, 85 min)		
			Mass loss /%	Penetration ratio /%	10°C Ductility /cm
58.5	39.2	80.3	0.08	75	30

Table 2
Physicochemical index of ATH.

Al_2O_3	Fe_2O_3	Na_2O	Burning loss	Moisture	Whiteness	PH value	Particle size
65.22 %	0.01 %	0.2 %	34.45 %	0.12 %	91.24 %	9.52	<15 μm

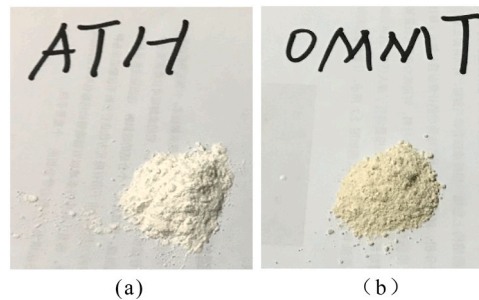


Fig. 2. Material images of (a) ATH and (b)OMMT.

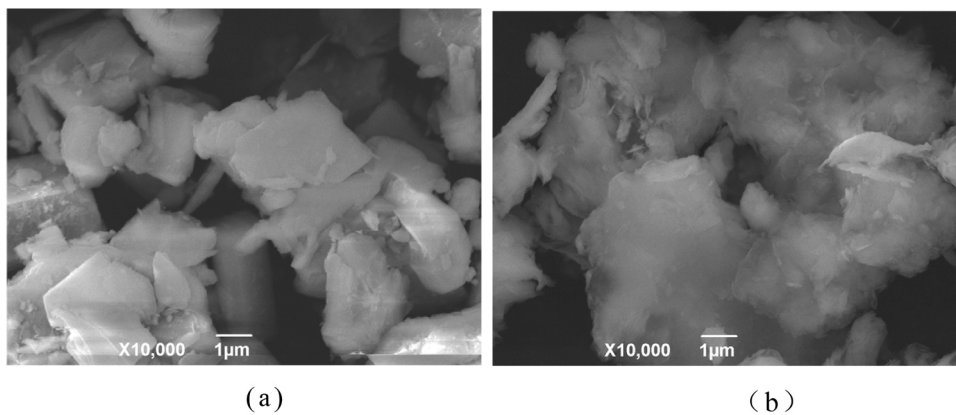


Fig. 3. SEM images of (a) ATH and (b)OMMT.

Based on this, the US Federal Highway Administration proposed to use MSCR test to evaluate the rutting resistance of asphalt materials. MSCR test selects different creep stress levels for continuous testing, and this loading mode can better simulate the response relationship between load and deformation of asphalt pavement. The evaluation indexes of MSCR test include creep recovery rate (R) and non-recoverable creep compliance (Jnr). Among them, the Jnr often has good correlation with the measured rut depth [22].

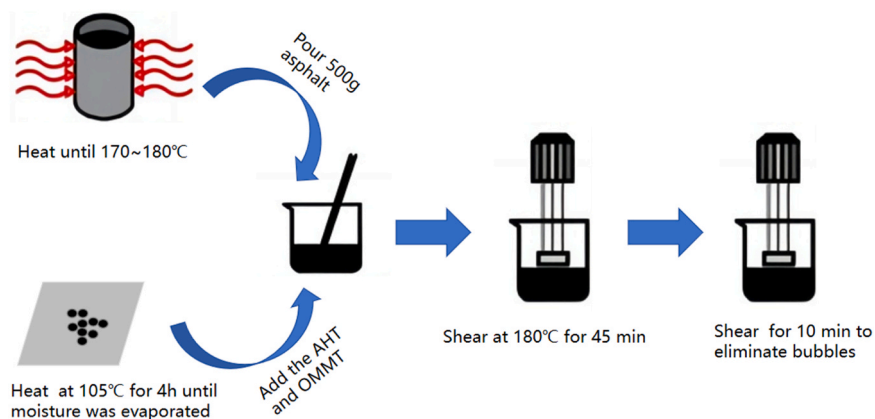


Fig. 4. The preparation method of ATH/OMMT/SBS modified asphalt.

The MSCR test was carried out according to the American AASHTOTP70–2010 standard. The creep stress level δ was selected as 0.1kPa and 3.2kPa respectively, and each stress level was tested continuously for 10 cycles. Each period included 1 s creep stress loading stage and 9 s stress unloading recovery stage. During this period, the cumulative strain curves of each cycle were recorded, and the total test time under the two stress levels was 200 s [23]. The test temperature was 64°C, the diameter of parallel plate was 25 mm, the distance between parallel plates was 1 mm, and the load frequency was 1.59 Hz.

The typical MSCR calculation process is shown in Fig. 5 The initial strain value at the beginning of each load cycle was recorded as ε_0 , the strain at the end of loading was recorded as ε_r , and the strain peak value of each loading cycle was recorded as ε_c . then the instantaneous shear strain ε_1 and non-recoverable strain in each cycle were calculated according to formula (1) and formula (2) [24]:

$$\text{Instantaneous shear strain } \varepsilon_1 = \varepsilon_c - \varepsilon_0 \quad (1)$$

$$\text{non-recoverable shear strain } \varepsilon_{10} = \varepsilon_r - \varepsilon_0 \quad (2)$$

The calculation process of the creep recovery rate (R) and non-recoverable creep compliance (Jnr) is shown in formula (3) and formula (4):

$$R = \sum \frac{\varepsilon_1 - \varepsilon_{10}}{\varepsilon_1} \times 100\% \quad (3)$$

$$J_{nr} = \sum \frac{\varepsilon_{10}}{\delta} \quad (4)$$

Creep recovery rate (R) and non-recoverable creep compliance (Jnr) are important indexes to evaluate the high temperature performance of asphalt materials. When the R value is larger and the Jnr value is smaller, it shows that the elastic deformation of modified asphalt accounts for a larger proportion of the total deformation under repeated creep load, that is, the stronger the ability of asphalt material to resist high temperature deformation, the better its high temperature performance. On the contrary, when the R value is smaller and the Jnr value is higher, it shows that the asphalt material has poor resistance to load and high temperature deformation, and its high temperature stability is also poor.

2.3.2. Linear amplitude sweep (LAS) test

The LAS test was conducted according to AASHTO TP-101-12 test procedure. LAS test is mainly divided into two steps [25]: Firstly, the sample was scanned in the frequency range of 0.2–100 Hz under 0.2 % strain to obtain the rheological parameters of asphalt materials. In this paper, the AR 1500EX rheometer of TA Instrument was used for testing. The test temperatures were 30°C, 45°C, 60°C and 75°C, respectively. The diameter of the parallel plate was 25 mm and the spacing was 1 mm. Secondly, the damage characteristics of asphalt materials were tested by linear amplitude strain scanning. The 8 mm parallel plate was selected, the spacing was 2 mm, the test temperature was 15°C, the strain control loading mode, the scanning time was 300 s. In order to quickly realized the fatigue damage process of asphalt materials, the amplitude of sine wave load loaded during the test increased linearly from 0.1 % to 30 %, as shown in Fig. 6.

In order to quantitatively characterized the cumulative fatigue damage degree of asphalt materials, the cumulative fatigue damage parameter D(t) was calculated by formula (5)–(6) [26,27].

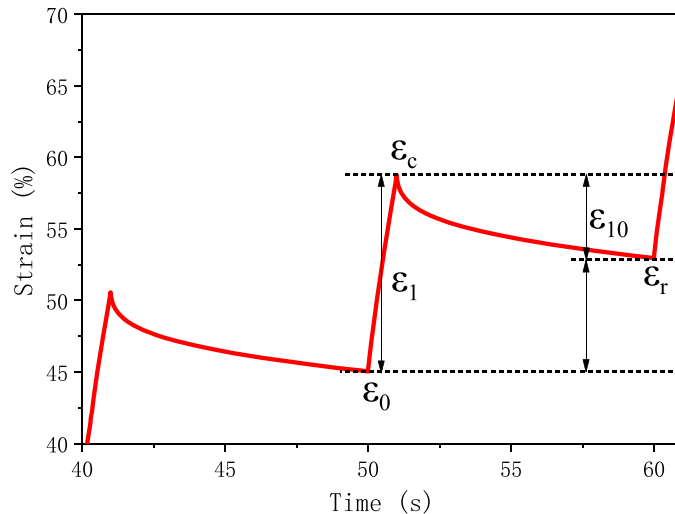


Fig. 5. MSCR shear test process.

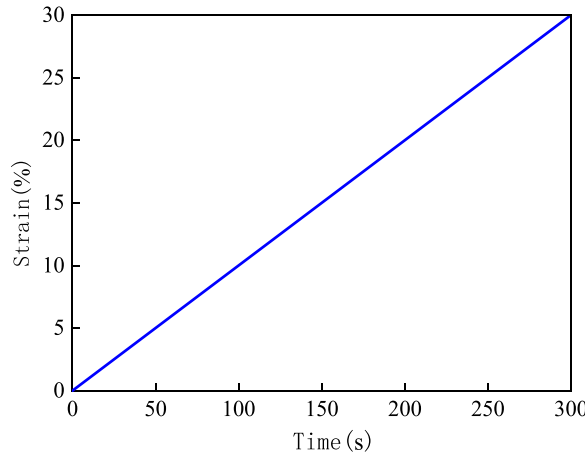


Fig. 6. Loading mode of LAS test.

$$D(t) \cong \sum_{i=1}^N [\pi \gamma_0^2 (C_{i-1} - C_i)] \frac{1}{(t_i - t_{i-1})^{1+\alpha}} \quad (5)$$

$$C_i = \frac{|G^*|(t)}{|G^*|_{initial}} \quad (6)$$

Where, γ_0 is the strain level, %; t is the strain action time, s; $|G^*|(t)$ is the complex shear modulus at time t , MPa; $|G^*|_{initial}$ is the initial scanning complex modulus, MPa; α is a coefficient, which is calculated by formula (7)–(8).

$$\log G'(\omega) = m(\log \omega) + b \quad (7)$$

$$\alpha = 1 + 1/m \quad (8)$$

Where, $G'(\omega)$ is the storage modulus, MPa; ω is the frequency, Hz; m is the linear fitting slope of the energy storage modulus and the logarithm of frequency.

The relationship between the integrity parameter C of asphalt material and the fatigue damage degree parameter D is shown in formula (9).

$$C(t) = C_0 - C_1(D)^{C_2} \quad (9)$$

Where, $C(t)$ is the degree of material damage; C_0 means that the material is not damaged, which is a fixed value of 1; C_1 and C_2 are the curve fitting parameters;

The fatigue failure point D_f is defined as the shear strain value corresponding to the maximum peak shear stress, as shown in formula (10). According to the viscoelastic continuous damage mechanics (VECD) model, calculate the fatigue life of asphalt material, as shown in formula (11):

$$D_f = \left(\frac{C_0 - C_p}{C_1} \right)^{\frac{1}{C_2}} \quad (10)$$

$$N_f = A(\gamma_{\max})^{-B} \quad (11)$$

Where, N_f is the fatigue life of asphalt material/time; f is the loading frequency (10 Hz); p is the stress peak, $\gamma_{\max}$ is the expected maximum strain level, %; A and B are the fatigue life coefficients, calculated according to formula (12):

$$A = \frac{f(D_f)^{(1+(1-C_2)\alpha)}}{(1 + (1 - C_2)\alpha) \times (\pi C_1 C_2)^\alpha}, \quad B = 2\alpha \quad (12)$$

2.3.3. Bending beam rheometer (BBR) test

The low temperature creep property of asphalt material plays an important role in the low temperature cracking performance of asphalt pavement. At present, bend beam rheometer (BBR) is widely used to evaluate the low temperature property of asphalt material. In this paper, the BBR test was carried out according to the AASHTOT313 code. During the test, the asphalt trabecula specimen (125 mm × 12.5 mm × 6.25 mm) was placed in anhydrous ethanol solution at the test temperature of -12 °C, -18 °C and -24 °C. A constant load (0.98 N) was applied in the middle of the trabecula supported by two fulcrums. The trabecula specimen was gradually

deformed under the load. Then the low temperature viscoelastic properties of asphalt materials were calculated by using the low temperature creep model, and the creep stiffness (S) and creep rate (m value) of asphalt materials were obtained. At the same time, the creep stiffness modulus (S) and creep rate (M value) of 60 s were taken to evaluate the low temperature creep characteristics of asphalt materials [28].

Creep stiffness (S) represents the ability of asphalt material to resist load deformation at low temperature, and the m value represents the sensitivity of creep stiffness to time and stress relaxation characteristics. The larger the S value of the asphalt material, the smaller the m value, which means that the brittleness of the asphalt material is greater, the sensitivity of the asphalt to stress is also greater, and the stress relaxation performance is poor. The smaller the S value, the larger the m value, indicating that the low temperature crack resistance of the asphalt material is better. In order to ensure the good low temperature performance of asphalt pavement, the AASHTO test regulations stipulate that the creep stiffness S value of the asphalt material should be smaller than 300 MPa, and the creep rate m value should not be less than 0.3 [29].

3. Results and discussion

3.1. MSCR test analysis of different modified asphalts

(1) Cumulative strain analysis of ATH/OMMT composite modified asphalt

The high-temperature creep characteristics of the composite modified asphalt material were analyzed by the MSCR method. Fig. 7. and Fig. 8. show the variation curves of the cumulative strain of different modified asphalts with time. It can be seen from the figure that under the action of different stress levels, the cumulative strains of all asphalts show similar variation laws. With the increase of action time, the cumulative strain increases gradually; the greater the shear stress is, the greater the cumulative strain value of modified asphalt is, and the cumulative strain value of 3.1kPa is significantly higher than that of 0.1kPa.

Fig. 7. shows the cumulative strain curve of single-doped modified asphalt with time. Under the action of different stress levels, the cumulative strain value of modified asphalt is as follows: ATH modified asphalt > control asphalt > OMMT-1 % > OMMT-3 %. The higher the cumulative strain value is, the worse the creep recovery performance of modified asphalt is and the worse its high temperature performance is. Compared with the control sample asphalt, when ATH modifier is added, the cumulative strain of ATH modified asphalt increases significantly. From the SEM image of ATH in Section 2.1, it can be seen that ATH modifier is tiny particles with smooth surface. When the temperature is high, it may be because ATH acts as a "lubricant" inside the modified asphalt, which reduces the frictional resistance that hinders the movement of asphalt molecules, and the asphalt material produces a large cumulative strain under the load, which significantly reduces the creep recovery performance of modified asphalt. This is different from Jia et al. 's conclusion in DSR test of ATH modified asphalt (the addition of ATH has a positive effect on high temperature performance (rutting factor)). It also shows that there are some defects in evaluating the high temperature performance of ATH modified asphalt with rutting factor index.

Compared with the control asphalt, the cumulative strain of OMMT modified asphalt is significantly reduced, and the greater the content of OMMT, the higher the degree of reduction, indicating that the improvement effect on the high temperature performance of the modified asphalt is also better. This may be due to the fact that the OMMT with lamellar structure forms an intercalated or exfoliated structure inside the modified asphalt, which constitutes a good three-dimensional network, which plays a good role in

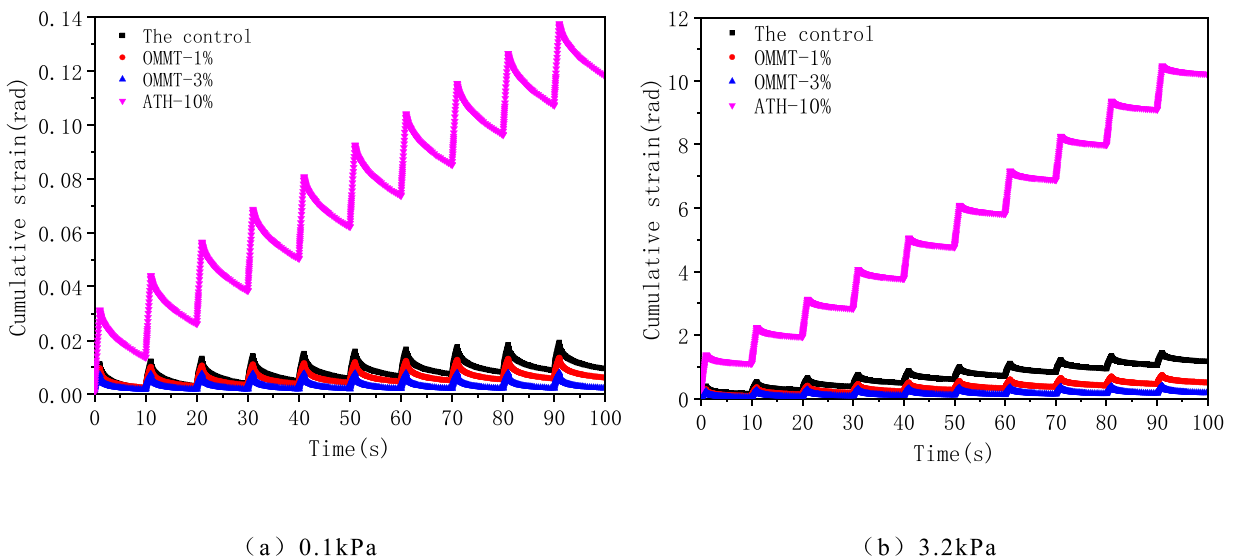


Fig. 7. Cumulative strain test results of single-mixed modified asphalt under different stresses.

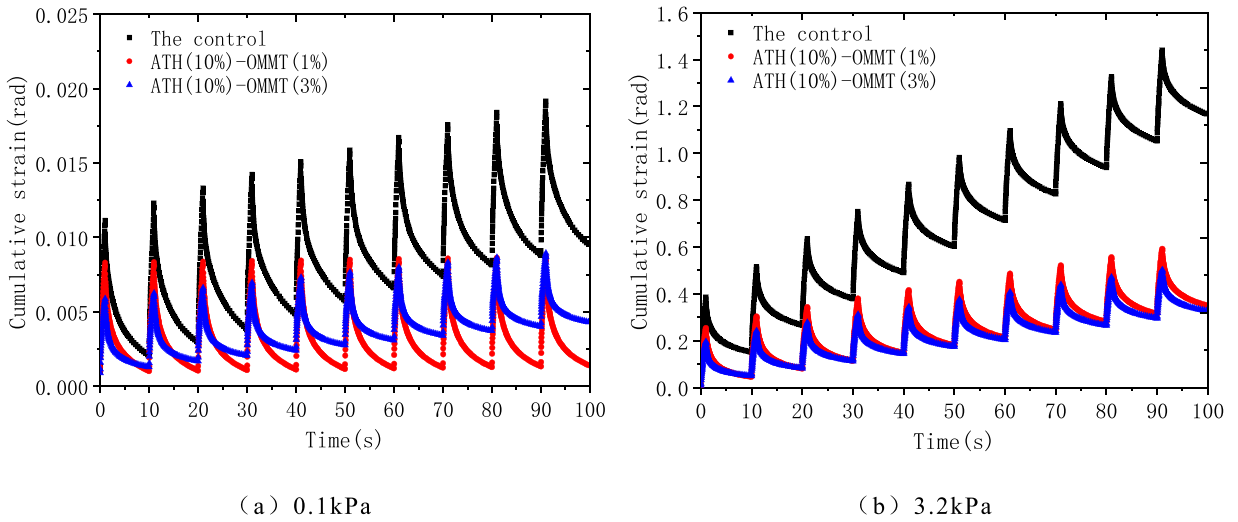


Fig. 8. Cumulative strain test results of composite modified asphalt under different stresses.

hindering the thermal motion between the asphalt molecules, thereby improving the high temperature performance.

Fig. 8 shows the variation curve of the cumulative strain of the composite modified asphalt with time. It can be seen from the figure that, compared with the control sample asphalt, the creep recovery performance of the modified asphalt is significantly improved, and the greater the load level is, the more obvious the improvement effect is. Combined with Fig. 7, it can be seen that the creep recovery performance of ATH single-doped modified asphalt is poor. When OMMT modifier is added to ATH modified asphalt, the repeated creep recovery performance of the two mixed modified asphalts is significantly improved. OMMT has a good improvement on the high temperature performance of ATH modified asphalt, so as to solve the engineering problems such as poor mechanical properties of ATH modified asphalt. E. Šušteršič et al. also showed that ATH and its participating composite modified asphalt have a significant effect on the high temperature performance of asphalt [30]. Meng Jia et al. also showed that the addition of OMMT has a positive effect on improving high temperature performance [31].

(2) Analysis of MSCR evaluation index of composite modified asphalt

The MSCR characteristic indexes of different modified asphalts were calculated according to formula (1)–(4), and the results are shown in Fig. 9.

Fig. 9(a). shows the calculated creep recovery rate (R) of all modified asphalts. The higher the R value, the stronger the elastic recovery deformation ability of the modified asphalt, and the better its high temperature performance. It can be seen from the diagram

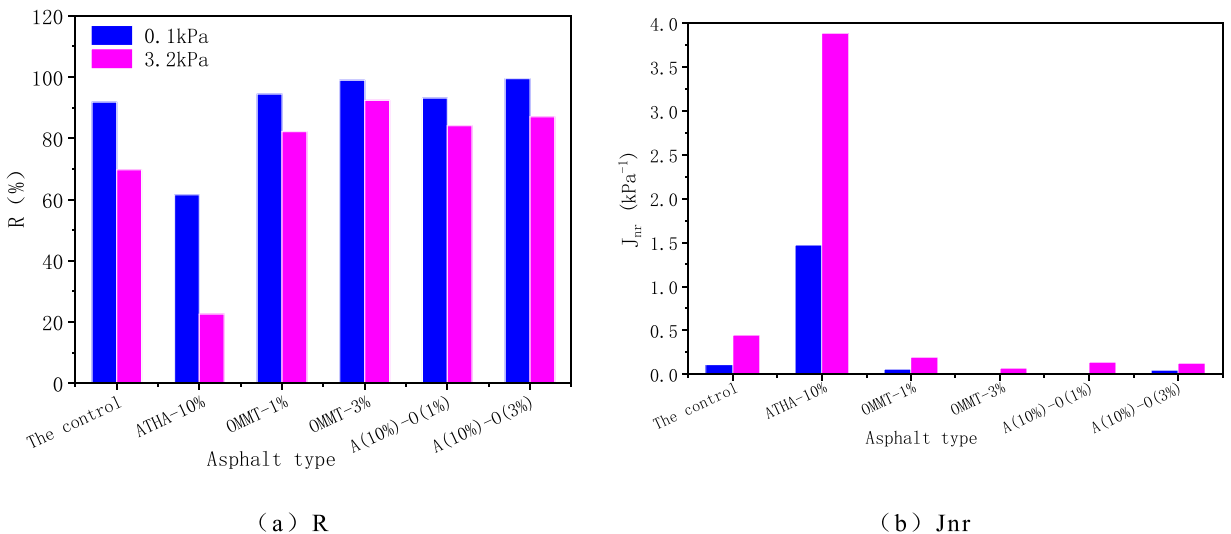


Fig. 9. Test results of R and J_{nr} of different modified asphalts.

that the change law of R value of all modified asphalt is the same under different stress, that is, the greater the load stress is, the smaller the R value is. At the same stress level, the R value of ATH in all modified asphalt is the smallest. At the level of 0.1kpa, its R value is reduced by 33.0 % compared with the control sample asphalt, and at the level of 3.2kpa, its R value is reduced by 67.6 %, indicating that ATH has a significant negative impact on the creep recovery rate of asphalt materials, and the creep recovery capacity of modified asphalt is the worst.

For the stress sensitivity of different asphalt materials, the change rate of creep recovery rate r of the control asphalt under different stress levels is 24.1 %. ATH increases the stress sensitivity of ATH modified asphalt, and the change rate of R value of ATH modified asphalt is 63.3 %. However, the change rate of R value of OMMT modified asphalt and composite modified asphalt is 15 %, indicating that OMMT modifier can significantly improve the high temperature performance of asphalt materials and reduce the stress sensitivity of modified asphalt at high temperature.

Compared with the control sample asphalt, the R value of ATH modified asphalt is significantly smaller, and its creep recovery ability is obviously worse, which may be caused by the lubrication effect of ATH modifier in the modified asphalt. When OMMT modifier is added to ATH modified asphalt, the R value of ATH/OMMT modified asphalt increases significantly, and the higher the stress level is, the more obvious the increasing effect is, which shows that OMMT modifier can improve the high temperature performance of ATH modified asphalt. By comparing the R values of different composite modified asphalts, it is found that the R value of ATH10 %-OMMT3 % is slightly higher than that of ATH10 %-OMMT1 % composite modified asphalt, but the improvement effect is not significant, indicating that the amount of OMMT modifier in the compound modified asphalt has little effect on its high temperature performance. The composite modified asphalt with good high temperature performance can be prepared by using a lower content of OMMT modifier.

By calculating the J_{nr} values of different modified asphalts, the results are shown in Fig. 9(b). As can be seen from the figure, contrary to the creep recovery rate R test results, the higher the stress level, the greater the non-recoverable compliance J_{nr} of the asphalt material. Compared with the control sample asphalt, the J_{nr} values of OMMT modified asphalt and composite modified asphalt are smaller, and at the same stress level, the higher the OMMT content is, the smaller the J_{nr} value of modified asphalt is, indicating that the permanent deformation of OMMT modified asphalt and composite modified asphalt under load is very small, and has good high temperature rutting resistance.

3.2. Analysis of medium temperature fatigue performance of different modified asphalts

3.2.1. Analysis of fatigue characteristics of different modified bitumen

In order to explore the fatigue performance of different modified asphalt, LAS test was used to test the fatigue life of modified asphalt at 15 °C. Fig. 10 shows the shear stress-shear strain curves of all modified asphalts, which can be used to analyze the dependence on strain and the degree of damage to asphalt materials during shear loading.

It can be seen from Fig. 10 that the shear stress-shear strain curves of all modified asphalts show similar changes. There is a peak value of shear stress in the process of LAS test, and the inflection point of shear stress is caused by the internal damage of asphalt samples caused by the increase of shear strain. Compared with the control sample asphalt, the maximum allowable strain corresponding to the peak stress of modified asphalt increases slightly with the addition of modifier. The maximum allowable strain of control sample asphalt is 16.08 %, and the maximum allowable strain of ATH and OMMT modified asphalt increases to 17.14 % and 17.62 %, indicating that the single modifier can improve the fatigue life of asphalt materials to a certain extent. The maximum allowable strain of ATH10 %-OMMT1 % and ATH10 %-OMMT1 % composite modified asphalt is 16.19 % and 16.47 % respectively. Compared with the single modified asphalt, the maximum allowable strain of composite modified asphalt is slightly smaller, indicating

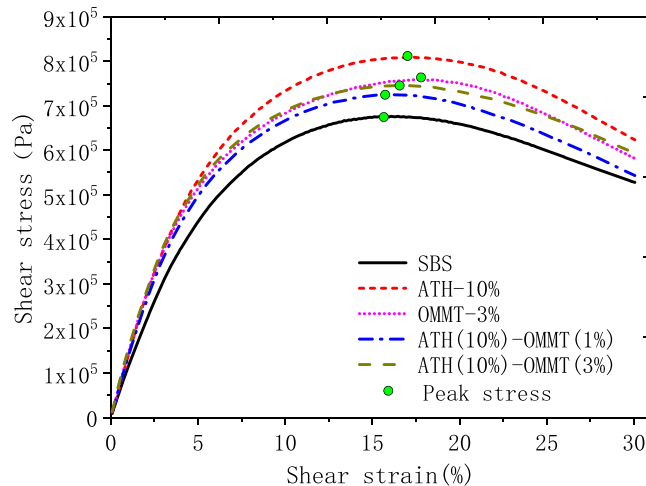


Fig. 10. Relationship between shear stress and strain of modified asphalt.

that the composite modified material has a negative effect on the fatigue life of asphalt.

In order to accurately describe the fatigue damage characteristics of modified asphalt during shear stress loading, the VECD model is used to draw the fatigue damage curve of asphalt samples as shown in Fig. 11, and the model parameters are shown in Table 3. The Abscissa D represents the cumulative damage parameters, and the ordinate C represents the integrity parameters of the asphalt sample. When $C = 1$, the asphalt sample is not damaged; when $C = 0$, the asphalt sample has been completely destroyed, and the reduction rate of C depends on the C1 and C2 values. The smaller the C1 and C2 values are, the greater the C value is.

As can be seen from Fig. 11, as the damage strength increases, the integrity parameters of the asphalt material also decrease gradually. Under the same damage strength, the integrity parameters of different modified asphalts are in the following order: ATH-10 % > control sample asphalt > OMMT-3 % > ATH10 %-OMMT1 % > ATH10 %-OMMT3 %. From Fig. 11 and Table 3, it can be seen that compared with the control asphalt, the reduction rate of the C value of ATH modified asphalt is the smallest, which is because the addition of ATH modifier significantly reduces the C1 value. The C1 value of single-mixed OMMT modified asphalt and ATH10 %-OMMT1 % composite modified asphalt slightly increased, indicating that the C value attenuation of the modified asphalt is higher than that of the control asphalt. However, the C1 value of the ATH10 %-OMMT3 % composite modified asphalt increased significantly, and the C value decay rate of the modified asphalt was significantly higher than that of the control asphalt, indicating that the internal ATH10 %-OMMT3 % composite modified asphalt under the action of shear stress. The degree of cumulative damage is much higher than that of the control asphalt, which affects its fatigue life.

3.2.2. Determination of fatigue life of different modified asphalts

The calculation results of the fatigue life coefficients A and B of different modified asphalts are shown in Table 3. According to formula (11), it can be seen that the higher the fatigue life coefficient A and the smaller the B value, the higher the fatigue life of the asphalt material. It can be seen from Table 3 that compared with the control sample asphalt, the fatigue life of modified asphalt depends on the comprehensive action of A value and B value, while ATH/OMMT compound modifier reduces A value and increases B value, which indicates that the compound modifier reduces the fatigue life of modified asphalt. Fig. 12 shows the fatigue life calculation results of different modified asphalts at 1 %, 2.5 %, 5 % and 10 % strain levels.

It can be seen from Fig. 12 that the fatigue life of ATH modified asphalt is significantly higher than that of other modified asphalt at different strain levels. At 2.5 % strain level, the fatigue life of ATH modified asphalt is 45,092, which is about 3 times higher than that of contrast sample asphalt, indicating that ATH modifier can significantly improve the fatigue life of modified asphalt materials. The effect of OMMT modifier on the fatigue life of modified asphalt varies with the strain level. When the strain level is lower (≤ 2.5 %), the fatigue life of OMMT modified asphalt is higher than that of control sample asphalt. Taking the fatigue life of 2.5 % strain level as an example, the fatigue life of control sample asphalt is 15,764, while that of OMMT modified asphalt is 23,222. It shows that OMMT modifier can improve the fatigue performance of modified asphalt at low strain level. However, when the strain level is higher (≥ 5 %), the fatigue life of OMMT modified asphalt is slightly lower than that of the control sample. The fatigue life of the two modified asphalts at 5 % strain level is 2002 (control sample) and 1953 (OMMT modified asphalt). Similarly, Shengjie et al. showed that the composite modification of OMMT and SBS showed higher milk fatigue performance in general, which is consistent with the conclusions in the low strain case in this study [31]. Respectively, indicating that OMMT modifier has a certain negative impact on the fatigue life of asphalt materials. In addition, at different shear strain levels, the fatigue life of the ATH/OMMT composite modified asphalt is lower than that of the control sample asphalt, indicating that the composite modifier has an adverse effect on the fatigue life of the modified asphalt.

In order to accurately predict the fatigue life of different modified asphalt, the logarithm of shear strain ϵ and the logarithm of fatigue life N_f were analyzed by linear fitting, and the fitting results are shown in Fig. 13 and Table 4. It can be seen from Table 4 that the correlation coefficients of the fitting variances are all 1, indicating that the prediction model has high accuracy, so that the fatigue

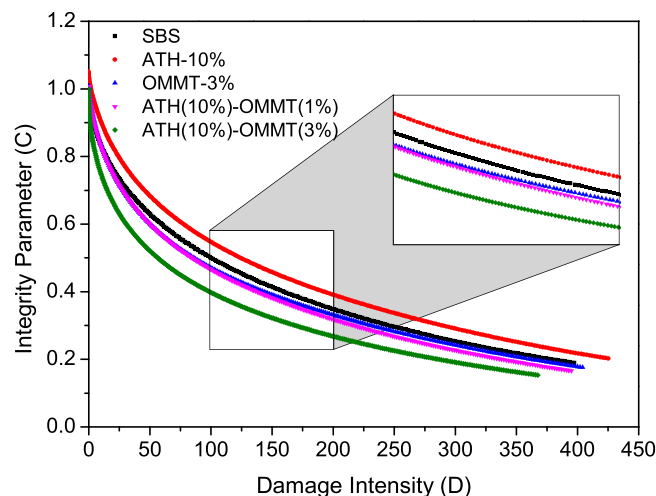
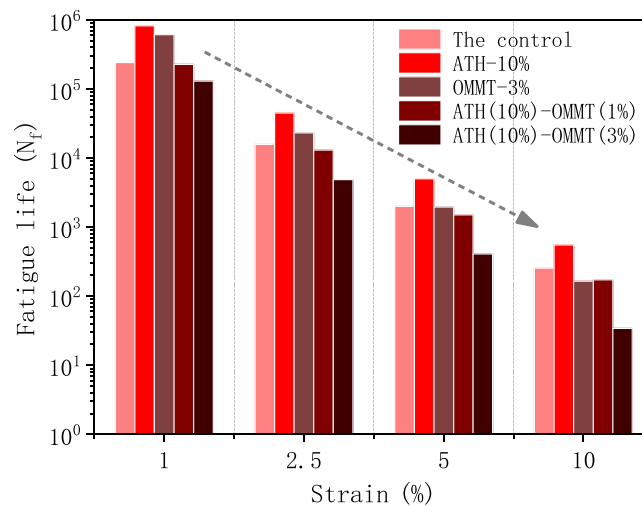


Fig. 11. Damage curve of modified asphalt under shear strain.

Table 3

Fitting results of different modified asphalt LAS test parameters.

Asphalt type	A	B	α	C_0	C_1	C_2	Failure strain (%)
The control	241,248.0	2.977	1.489	1.0	0.071	0.417	16.08
ATH-10 %	826,232.9	3.174	1.587		0.041	0.506	17.14
OMMT-3 %	612,850.3	3.572	1.786		0.081	0.397	17.62
ATH10 %-OMMT1 %	227,634.8	3.121	1.561		0.079	0.406	16.19
ATH10 %-OMMT3 %	130,208.3	3.581	1.791		0.133	0.320	16.47

**Fig. 12.** Calculation results of fatigue life of different modified asphalts.

life of different modified asphalts at a temperature of 15 °C can be accurately predicted.

In addition, by comparing the fatigue life linear fitting results of Fig. 14 and Table 4, it is found that the slope of the fatigue life fitting equation of different modified asphalt is quite different. The greater the absolute value of the slope of the fitting equation, the stronger the sensitivity of the fatigue life of asphalt materials to the strain level, that is, the poor ability of asphalt pavement to resist fatigue cracking under different loads. The smaller the absolute value of the slope, the weaker the sensitivity of the fatigue life of the asphalt material to the strain level, and the better the stability of the fatigue life.

It can be seen from Table 4 that the absolute value of the slope of the fatigue life fitting equation of all modified bitumen is larger than that of the control sample asphalt (2.977), and their growth rates are ATH modified asphalt (6.6 %), OMMT-3 % (20.0 %), ATH10 %-OMMT1% (4.8 %) and ATH10 %-OMMT3 % (20.3 %), respectively. ATH modifier has little effect on the fatigue life stability of asphalt materials, and OMMT modifier is the main factor affecting the fatigue life stability of composite modified asphalt. The higher the content of OMMT, the greater the decrease of fatigue life stability. Therefore, the fatigue life of OMMT-3 % and ATH10 %-OMMT3 % modified asphalt is more sensitive to the change of load, and the stability of fatigue life is poor.

3.3. Analysis of low temperature rheological properties of different modified asphalts

Fig. 10 shows the deflection curve of trabecular specimens of different modified asphalt materials during the BBR test (take −12°C as an example). The smaller the deflection value is, the smaller the deformation of the asphalt material is in the process of loading, and its flexibility is poor, so it is prone to brittle fracture. The larger the deflection value is, the greater the deformation of asphalt material is in the process of loading, and the better the low temperature crack resistance of asphalt material is.

It can be seen from Fig. 14 that the deformation of asphalt materials increases gradually with the increase of loading time, but the deformation curves of different modified asphalt materials are also quite different. Under the same loading time, the deflection values of different modified asphalts are in the following order: comparative sample asphalt > OMMT-3 % > ATH10 %-OMMT1 % > OMMT-1 % > ATH10 %OMMT3 % > ATH-10 %, that is, the low temperature crack resistance of different modified asphalts becomes worse.

Compared with the control sample, the addition of ATH modifier significantly reduces the deflection deformation value of the asphalt material, indicating that the brittleness of the asphalt material increases and the low temperature deformation resistance is significantly reduced. For OMMT modified asphalt, although the addition of OMMT modifier reduces the deflection deformation of the modified asphalt, with the increase of its content, the deflection deformation of the modified asphalt increases significantly. The deflection deformation of OMMT-3 % modified asphalt is significantly higher than that of OMMT-1 %.

Compared with ATH modified asphalt, due to the addition of OMMT modifier, the low temperature deformation ability of ATH/OMMT composite modified asphalt is significantly improved, and the deflection deformation of composite modified asphalt is

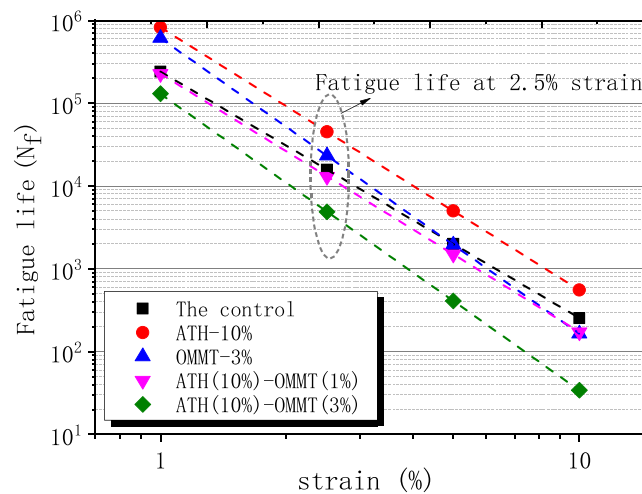


Fig. 13. Logarithmic linear fitting curve of fatigue life and strain of modified asphalt.

Table 4

Fatigue life prediction equation of different modified asphalts.

Asphalt type	Shear strain (%)	Fatigue life N_f	Estimation equation	Correlation coefficient R^2
The control	1	241,248	$\lg N_f = -2.977 \lg \varepsilon + 5.382$	1
	2.5	15,764		
	5	2002		
	10	254		
ATH-10 %	1	826,233	$\lg N_f = -3.173 \lg \varepsilon + 5.917$	1
	2.5	45,093		
	5	4997		
	10	554		
OMMT-3 %	1	612,850	$\lg N_f = -3.572 \lg \varepsilon + 5.787$	1
	2.5	23,222		
	5	1953		
	10	164		
ATH10 %-OMMT1 %	1	227,635	$\lg N_f = -3.121 \lg \varepsilon + 5.357$	1
	2.5	13,039		
	5	1499		
	10	172		
ATH10 %-OMMT3 %	1	130,208	$\lg N_f = -3.581 \lg \varepsilon + 5.114$	1
	2.5	4892		
	5	409		
	10	34		

significantly higher than that of ATH modified asphalt. However, the deflection deformation of composite modified asphalt decreases with the increase of OMMT content, and the deflection deformation of ATH10 %-OMMT1 % composite modified asphalt is significantly higher than that of ATH10 %-OMMT3 % modified asphalt. Therefore, in order to improve the low temperature cracking resistance of the composite modified asphalt, the content of OMMT in the composite modified asphalt should not be too high.

The low temperature creep stiffness modulus S and creep rate m of different modified asphalt materials were calculated, and the results are shown in Fig. 15. It can be seen from Fig. 15 that the lower the test temperature is, the greater the difference of stiffness modulus S and m between different modified asphalt materials is. When the test temperature is -12°C and -18°C , the S and m values of modified asphalt are almost the same as those of contrast sample asphalt, and the changes of S and m values of different modified asphalts are the most obvious at -24°C . The S value of all modified asphalt at -18°C is less than 300 MPa, and the m value is greater than 0.30, that is, the low temperature grade is -18°C , indicating that adding modifiers will not affect the low temperature PG grade of asphalt materials, but will affect the flexibility of asphalt materials, and the lower the temperature is, the greater the influence degree is.

As can be seen from Fig. 15 (a), the stiffness modulus S of the contrast sample asphalt is the smallest, indicating that the low temperature flexibility of the asphalt is the best. The S value of ATH modified asphalt is the largest, indicating that ATH modifier has a large negative impact on the low temperature performance of asphalt materials. Compared with the control asphalt, the S value of OMMT-1 % modified asphalt has a larger increase, but the S value of OMMT-3 % modified asphalt is close to that of the control asphalt. It shows that adding 3 % OMMT has no significant effect on the low temperature flexibility of asphalt materials. The reason may be that when the content of OMMT is small, a good network structure is not formed inside the modified asphalt, while when the content of

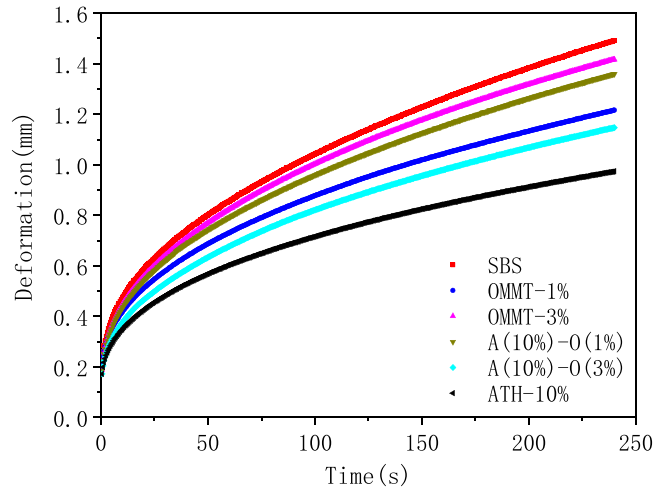


Fig. 14. BBR test curves of different modified asphalts (-12°C).

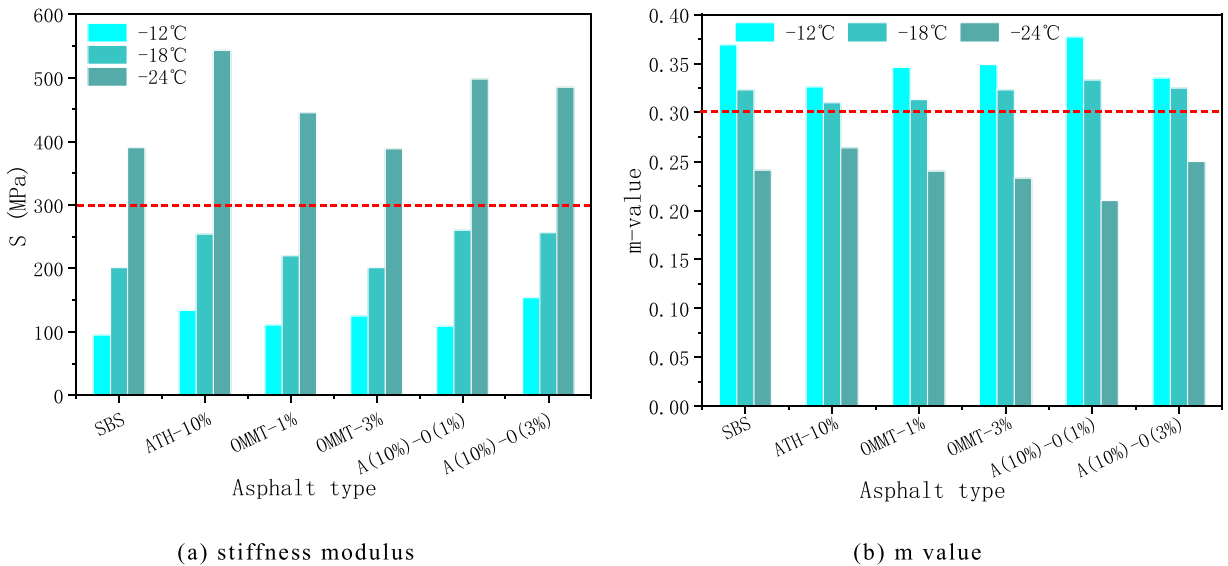


Fig. 15. Test results of stiffness modulus S and m of different modified asphalts.

OMMT is 3 %, The OMMT in the modified asphalt is uniformly distributed and forms a good three-dimensional network structure. For composite modified asphalt, the S value of ATH10 %-OMMT1 % and ATH10 %-OMMT3 % composite modified asphalt is significantly lower than that of ATH modified asphalt, indicating that OMMT modifier can improve the low temperature flexibility of ATH modified asphalt to a certain extent. Similarly, Shengjie et al. showed that OMMT had a certain effect on improving the crack resistance, but the effect of its content factor was not significant [31].

The test results of creep rate m of modified asphalt are similar to those of stiffness modulus S . As shown in Fig. 15 (b), the m value of the control sample asphalt is the largest and its low temperature performance is the best; the addition of ATH modifier reduces the m value of asphalt material, and OMMT modifier also reduces its m value, and the content of OMMT has little effect on the m value. Compared with ATH modified asphalt, the m value of composite modified asphalt increases slightly, which further proves that OMMT can improve the low temperature of ATH modified asphalt. The m value of ATH10 %-OMMT1 % is slightly higher than that of ATH10 %-OMMT3 % modified asphalt, and the content of OMMT in composite modified asphalt should not be too high. The reason may be that when the content of OMMT is large, it is easy to agglomerate with ATH modifier, which leads to the increase of non-uniformity of the system and the phenomenon of stress concentration under load, which affects its low temperature performance.

4. Conclusions

In this paper, the high-temperature rheological properties, medium-temperature fatigue properties and low-temperature crack

resistance of ATH / OMMT composite modified asphalt were systematically analyzed, and the influence of composite modified materials on the rheological properties of asphalt was discussed. The main conclusions are as follows :

- (1) Hydroxide aluminum (ATH) reduces the repeated creep recovery ability of asphalt materials, while modified asphalt with organic montmorillonite (OMMT) demonstrates favorable creep recovery performance. The higher the content of OMMT, the greater the improvement in creep recovery performance of the modified asphalt. Moreover, the high-temperature creep recovery performance of the composite modified asphalt is significantly better than that of the control sample asphalt and ATH-modified asphalt.
- (2) Sole addition of the modifier can significantly enhance the fatigue life of the asphalt material, but under high-strain conditions, OMMT modifier will decrease the material's fatigue life. Composite modification agents will reduce the fatigue life of the modified asphalt.
- (3) The modifiers do not alter the low-temperature PG grade of the asphalt material, but their flexibility worsens as the temperature decreases. The sole addition of ATH modifier has a significant negative impact on the low-temperature performance of the asphalt material, yet incorporating OMMT modifier provides some improvement in the low-temperature flexibility of the ATH-modified asphalt.
- (4) The research results indicate that the combined modification of asphalt with ATH and OMMT is a mutually beneficial material integration for the rheological properties of asphalt. This suggests that in the future, it can be applied as a tunnel fire-retardant and smoke-suppressing material with good mechanical properties for asphalt pavement.

CRedit authorship contribution statement

Xinyi He: Writing - original draft. **Ming Tan:** Funding acquisition. **Xiaolin Zhu:** Writing – review & editing. **Shengrui Tan:** Formal analysis, Data curation. **Xiaolong Yang:** Methodology, Investigation, Project administration.

Declaration of Competing Interest

No potential conflict of interest was reported by the authors.

Data availability

The data that has been used is confidential.

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